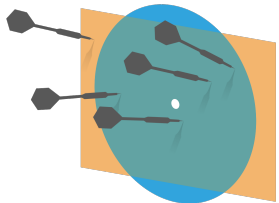


Probability and Statistics for Computer Science



Credit: wikipedia

Conditional probability comes back in matrix!

Last time

- ✱ Spectral Clustering (II)
- ✱ Markov Chain (I)

Objectives

- ✱ Markov Chain (II)

Markov chain in terms of probability

- ✱ Let $X_0, X_1, \dots$ be a sequence of discrete finite-valued random variables
- ✱ The sequence is a Markov chain if the probability distribution X_t only depends on the distribution of the immediately preceding random variable X_{t-1}

$$P(X_t | X_0, \dots, X_{t-1}) = P(X_t | X_{t-1})$$

- ✱ If the conditional probabilities (transition probabilities) do **NOT** change with time, it's called **constant Markov chain**.

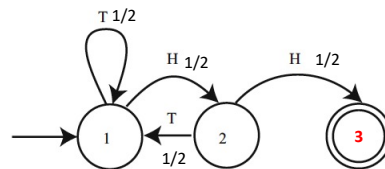
$$P(X_t | X_{t-1}) = P(X_{t-1} | X_{t-2}) = \dots = P(X_1 | X_0) = C$$

a const. matrix

Coin example

- ✱ Toss a fair coin until you see two heads in a row and then stop, what is the probability of stopping after exactly n flips?
- ✱ Use a state diagram, which is a **directed graph**. Circles are the states of likely outcomes. Arrow directions show the direction of transitions. Numbers over the arrows show transition probabilities.

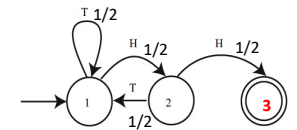
1 -> Start or just had tail/restart
 2 -> had one head after start/restart
 3 -> 2 heads in a row/Stop



The model helps form recurrence formula

- ✱ Let p_n be the probability of stopping after n flips

$$p_1 = 0 \quad p_2 = 1/4 \quad p_3 = 1/8 \quad p_4 = 1/8 \quad \dots$$



The model helps form recurrence formula

- ✱ Let p_n be the probability of stopping after n flips

$$p_1 = 0 \quad p_2 = 1/4 \quad p_3 = 1/8 \quad p_4 = 1/8 \quad \dots$$

- ✱ If $n > 2$, there are two ways the sequence starts

- ✱ Toss T and finish in $n-1$ tosses
- ✱ Or toss HT and finish in $n-2$ tosses

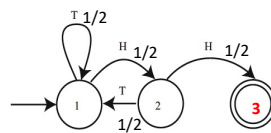
- ✱ So, we can derive a recurrence relation

$$p_n = \frac{1}{2}p_{n-1} + \frac{1}{4}p_{n-2}$$

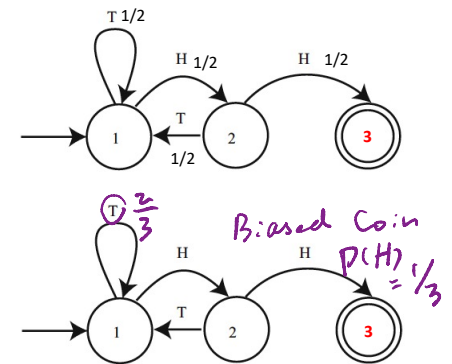
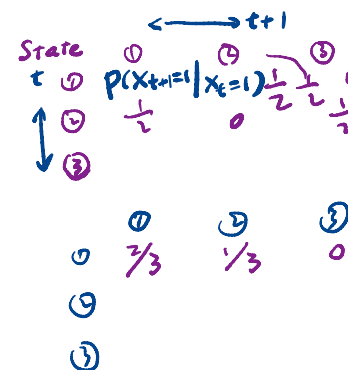
$P(T)$

$P(HT)$

"TH" + $n-2$ tosses
 is counted already.

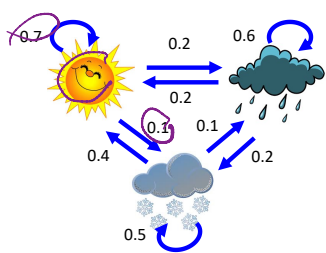


Transition probability btw states



Transition probability matrix: weather model

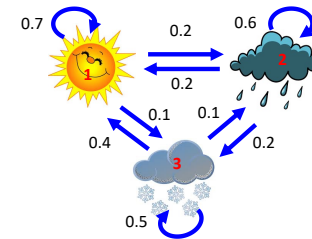
- Let's model daily weather as one of the three states (Sunny, Rainy, and Snowy) with Markov chain that has the transition probabilities as shown here.



	Sun	Su	Rain
Sun	0.7	0.1	0.2
Su	0.4	0.5	0.1
Rain	0.2	0.2	0.6

Transition probability matrix: weather model

- Let's model daily weather as one of the three states (Sunny, Rainy, and Snowy) with Markov chain that has the transition probabilities as shown here.



i , the current state at time point t
 j , the next state at time point $t+1$

	Sunny	Rainy	Snowy
Sunny	0.7	0.2	0.1
Rainy	0.2	0.6	0.2
Snowy	0.4	0.1	0.5

$P =$

The transition probability matrix

Q: Is this TRUE?

For a constant Markov Chain, at any step t , the probability distribution among the states remain the same.

transition matrix P is constant.
 not $P(X_t)$

A. Yes.

☒ B. No.

Q: The transition probabilities for a node sum to 1

A. Yes.

☒ B. No.

Only outgoing edges' sum to 1

Transition probability matrix properties

- ✱ The transition probability matrix $\mathbf{P}$ is a square matrix with entries p_{ij}

- ✱ Since $p_{ij} = P(X_t = j | X_{t-1} = i)$

$$p_{ij} \geq 0 \quad \text{and} \quad \sum_j p_{ij} = 1$$

$$\mathbf{P} = \begin{array}{ccc|l} \text{Sunny} & \text{Rainy} & \text{Snowy} & \\ \hline 0.7 & 0.2 & 0.1 & \text{Sunny} \\ 0.2 & 0.6 & 0.2 & \text{Rainy} \\ 0.4 & 0.1 & 0.5 & \text{Snowy} \end{array}$$

The transition probability matrix

Probability distributions over states

- ✱ Let $\boldsymbol{\pi}$ be a row vector containing the probability distribution over all the finite discrete states at $t=0$

$$\pi_i = P(X_0 = i)$$

- ✱ For example: if it is rainy today, and today is $t=0$, then

$$\boldsymbol{\pi} = [0 \quad 1 \quad 0]$$

- ✱ Let $\mathbf{p}^{(t)}$ be a row vector containing the probability distribution over states at time point t

$$p_i^{(t)} = P(X_t = i)$$

Propagating the probability distribution

- ✱ Propagating from $t=0$ to $t=1$,

$$\begin{aligned} P_j^{(1)} &= P(X_1 = j) \\ &= \sum_i P(X_1 = j, X_0 = i) \\ &= \sum_i P(X_1 = j | X_0 = i) P(X_0 = i) \\ &= \sum_i p_{ij} \pi_i \end{aligned}$$

- ✱ In matrix notation,

$$\mathbf{p}^{(1)} = \boldsymbol{\pi} \mathbf{P}$$

$$p^{(1)} = \pi p^2$$

Probability distributions:

- ✱ Suppose that it is rainy, we have the initial probability distribution.

$$\boldsymbol{\pi} = [0 \quad 1 \quad 0]$$

- ✱ What are the probability distributions for tomorrow and the day after tomorrow?

$$\mathbf{p}^{(1)} = \boldsymbol{\pi} \mathbf{P}$$

$$\mathbf{p}^{(2)} = \mathbf{p}^{(1)} \mathbf{P}$$

Propagating to $t = \infty$

- We have just seen that

$$p^{(2)} = p^{(1)}P = (\pi P)P = \pi P^2$$

- So, in general $p^{(t)} = \pi P^t$

- If one state can be reached from any other state in the graph, the Markov chain is called **irreducible (single chain)**.

- Furthermore, if it satisfies: $\lim_{t \rightarrow \infty} \pi P^t = S$

then the Markov chain is stationary and S is the stationary distribution.

Stationary distribution

- The stationary distribution S has the following property:

$$sP = s$$

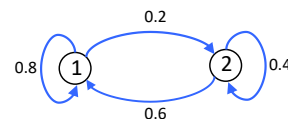
- S is a row eigenvector of P with eigenvalue 1

$$S = \lim_{t \rightarrow \infty} \pi \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.4 & 0.1 & 0.5 \end{bmatrix}^t = \left[\frac{18}{37} \quad \frac{11}{37} \quad \frac{8}{37} \right]$$

Example: Up-to-date or behind model

State 1: Up-to-date

State 2: Behind



What's the transition matrix?

If I start with $\pi = [0, 1]$, what is my probability of being up-to-date eventually? $3/4$

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$$

$$\lim_{t \rightarrow \infty} p(X_t) = S \quad SP = S$$

Example: Up-to-date or behind model

$$SP = S \Rightarrow (SP)^T = S^T \Rightarrow P^T S^T = S^T$$

$$\text{Let } A = P^T; \quad S^T = u$$

$$A = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix}$$

$$\text{Given } u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$Au = u$$

$$u_1 + u_2 = 1$$

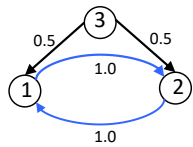
$$|A - I| = 0$$

$$\text{from (1)} \Rightarrow -0.2 * u_1 + 0.6 * u_2 = 0$$

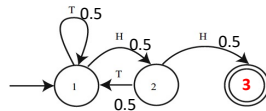
$$\begin{vmatrix} -0.2 & 0.6 \\ 0.2 & -0.6 \end{vmatrix} u = 0 \quad (1)$$

$$\Rightarrow \begin{cases} u_1 = 0.75 \\ u_2 = 0.25 \end{cases}$$

Examples of non-stationary Markov chains



Periodic

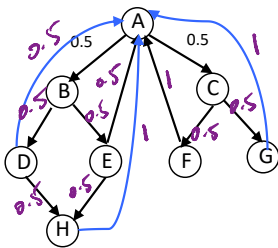


Absorbing

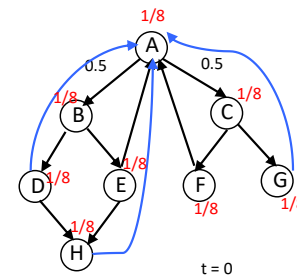
PageRank Example

- How to rate web pages objectively?
- The PageRank algorithm by Page et al. made **Google** successful
- The method utilized **Markov chain model** and applied it to the large list of webpages.
- To illustrate the point, we use a small-size example and assume a simple **stationary model**.

Suppose we are randomly surfing a network of webpages



Initialize the distribution uniformly

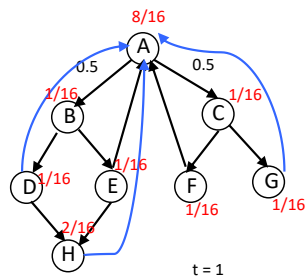


$$\pi = [1/8 \ 1/8 \ 1/8 \ 1/8 \ 1/8 \ 1/8 \ 1/8 \ 1/8]$$

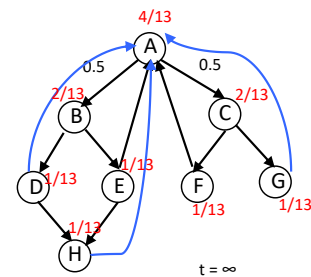
$$P = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

in πP^t
 $t \rightarrow \infty$

Update the distribution iteratively



Until the stationary distribution



D

$L(D|\theta)$

θ

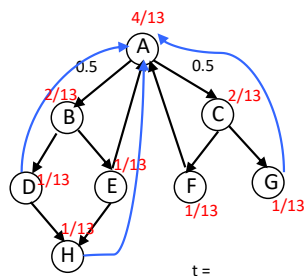
$\hat{\theta}$

MLE

$p(\theta|D)$

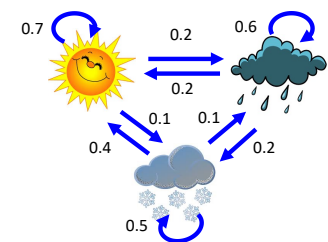
If the surfer get trapped

- Allow "teleport" with small probability from any page to another
- Or allow "teleport" with user input of URL

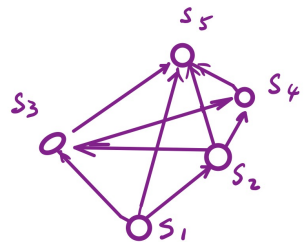


Diverse applications of Markov Model

- Communication network
- Queue modeling
- DNA sequence modeling
- Natural language processing
- Single-cell large data analysis
- Financial/Economic model
- Music



Communication network example



	S_1	S_2	S_3	S_4	S_5
S_1	0.1	0.2	0.4	0	0.3
S_2	0	0.1	0.1	0.2	0.6
S_3	0	0	0.1	0.1	0.8
S_4	0	0	0	0.1	0.9
S_5	0	0	0	0	1.0

$$P_{SS}^{(n)} = \pi_1 \cdot P[C, S]$$

(1x5) 5x1

$$\pi_0 = [1 \ 0 \ 0 \ 0 \ 0]$$

$$P_{SS}^{(n)} = ?$$

$$P_{SS}^{(1)} =$$

$$P_{SS}^{(2)} = ?$$

Markov model of chords

	F	C	G
F		0.4	0.1
C		0.2	0.7

$$P(F \rightarrow G \rightarrow C)$$

$$\text{Assume } X_0 = F \quad G \quad 0.2 \quad 0.4 \quad 0.4$$

$$P(X_0 = F, X_1 = G, X_2 = C) = ?$$

$$\pi_0 = [1 \ 0 \ 0]$$

$$P(X_2 = C | X_0 = F, X_1 = G) = P(X_0 = F, X_1 = G)$$

Joint probability

$$P(X_2 | X_0, X_1) = P(X_2 | X_1)$$

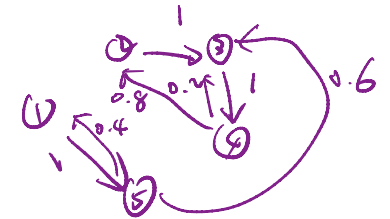
Markov
Property

$$P(X_2 = C | X_1 = G) = 0.4$$

$$\begin{aligned} P(X_2 = C, X_1 = G, X_0 = F) \\ &= 0.4 \cdot P(X_1 = G, X_0 = F) \\ &= 0.4 \cdot \frac{P(X_1 | X_0) P(X_0)}{P(F)} = 1 \\ &= 0.2 \end{aligned}$$

Probability over Markov Chain

- Draw the state diagram, how many 4 step paths are there that begin at state 1?



$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 1.0 \\ 0 & 0 & 1.0 & 0 & 0 \\ 0 & 0 & 0 & 1.0 & 0 \\ 0 & 0.8 & 0.2 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 & 0 \end{pmatrix}$$

Final Exam

- ⌘ Time: 8am 5/13 Wed. Central Time
- ⌘ Duration: 3hrs
- ⌘ Content coverage: Ch1-14, except 8, details are on Canvas
- ⌘ Open book and lecture notes
- ⌘ Format: 50 multiple choices, on PrairieTest proctored by Staff in classrooms (CIF0027 and CIF 4031)

Assignments

- ⌘ Finish Chapter 12 of the textbook
- ⌘ Project

Additional References

- ⌘ Robert V. Hogg, Elliot A. Tanis and Dale L. Zimmerman.
“Probability and Statistical Inference”
- ⌘ Kelvin Murphy, “Machine learning, A Probabilistic perspective”

